

the SDK. The preferred way to install scratchbox and SDK is to use the automated installation scripts, though manual installation steps are also provided. Detailed instructions are found at http://wiki.maemo.org/Documentation/Maemo_5_Final_SDK_Installation. In short, the procedure for Debian-based systems is as follows:

- Install Maemo SDK build environment.
- Set up and create development targets.
- Start developing.

Another very good source of information on setting the development environment can be found at <http://maemo-sdk.garage.maemo.org/install-ubuntu.html>.

Getting started with the development

With the SDK installed and working in place, you can test the development environment. Create the usual simple command line-based Hello World application in C:

```
#include <stdio.h>
int main(int argc, char** argv)
{
    printf("Hello world!\n");
    return 0;
}
```

Save the file as *sample.c* in the root of your development environment. Remember you should compile this program under the SDK and not your normal system. Unless you build the program under the SDK, you'll not not whether the SDK works or not.

Now test the application by issuing the following command:

```
[aditya@archie: ~] > gcc -Wall -g sample.c -o sample
[aditya@archie: ~] > ls sample*
sample sample.c
```

GCC's `-g` flag is used to add debugging symbols to the output file. The `-Wall` flag will instruct the compiler to enable most of the syntax and other warnings generated on compilation. `-o sample` means the output binary filename is 'sample'. Now run the program (`./sample`) and you shall see 'Hello World' printed on the terminal. Congratulations, it means your environment is up and running.

A step further: diving into the GUI

It's time now for some graphical sorcery. A simple GTK+ based window is demonstrated in the sample program below. The program is taken from the official Maemo Documentation and is released by Nokia under the CC license.

```
#include <stdio.h> /* EGT, */
/* Introduce types and prototypes of GTK+ for the compiler */
#include <gtk/gtk.h>
int main(int argc, char** argv)
{
```

Scratchbox

Maemo development is done in a Sandbox environment. Scratchbox delivers this sandbox environment on your system, providing development tools and at the same time, isolating the rest of the system from itself. Since there's a virtual environment available for development, cross-compiling is very easy.

The name 'Scratchbox' is a derivation from 'Linux from Scratch' + 'chroot jail' (sandbox). Scratchbox uses the Linux feature of changing the path of a file that a process can see. It changes the root (`/`) of the system to its own directory (same thing done in the case of a jail shell). This is the isolated area where all the work is done—a sandbox. Apart from directory switching, diversion of system calls, library references and other small changes makes development easier.

So, basically the Scratchbox is:

- a process of implementation of an isolated sandbox environment.
- a collection of tools to make cross-compilation and development easier.
- a collaborative environment supporting multiple developers with individual profile supports.
- a mechanism called `sbrsh` that supports execution of the binaries directly on the target system.
- a cross-execution simulator generally implemented through QEMU.

But Scratchbox does not end here. You can develop your own applications under the Scratchbox environment. Debian users will get a favour here as well. It includes some integration of Debian package management so that once the source files are set up correctly, all you need is to write a couple of configuration files and you can generate binary packages for various architectures. There are many other features integrated in Scratchbox and you can only enjoy it fully when you know them all.

```
/* We'll have two references to two GTK+ widgets */
GtkWindow* window;
GtkLabel* label;
/* Initialize the GTK+ library */
gtk_init(&argc, &argv);

/* Create a window with window border width of 12 pixels and a
title text. */
window = g_object_new(GTK_TYPE_WINDOW,
    "border-width", 12,
    "title", "Hello GTK+",
    NULL);

/* Create the label widget */
label = g_object_new(GTK_TYPE_LABEL,
    "label", "Hello World",
    NULL);
/* Pack the label into the window layout. */
gtk_container_add(GTK_CONTAINER(window), GTK_WIDGET(label));
```